

Computer Networks

1. BOOK: Computer Networks
By Andrew Tannenbaum 5th Edition
2. Data Communications and Networks:
3rd ed by Behrouz A. Forouzan. Tata
McGraw Hill Publication.

Facilitator: Dr. Rashmi Welekar

Computer Networks

7 layers Of OSI Model

Computer Networks

Course Learning Objective:

- 1.To develop an understanding of modern network architectures from a design and performance perspective.
- 2.To introduce the student to the major concepts involved in network protocols.
3. To provide an opportunity to do network programming

Computer Networks

Course Outcomes:

1. Understand basics of computer networks and reference models
2. Identify the Design issues of each layer of OSI model
3. Implement the protocols of OSI model

Introduction

3 W of Computer Networks

WHY
WHAT
WHEN

WHY

- Computer networks are becoming pervasive
- Virtually all software is networked
- every developer must know the basis of networking
- Internet is used daily by millions (if not billions of people)
 - Web
 - Email
 - Chat
 - Skype
 - Movies / MP3s
- ... and it is getting bigger and bigger

WHAT

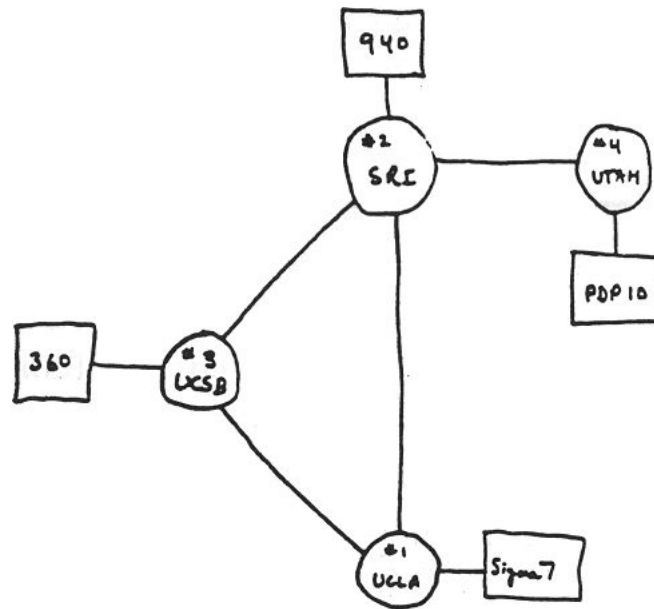
Interconnected collection of autonomous computers.

WHEN

SEPT 1969 ARPANET: 2 nodes connected

WHEN

DECEMBER 1969:



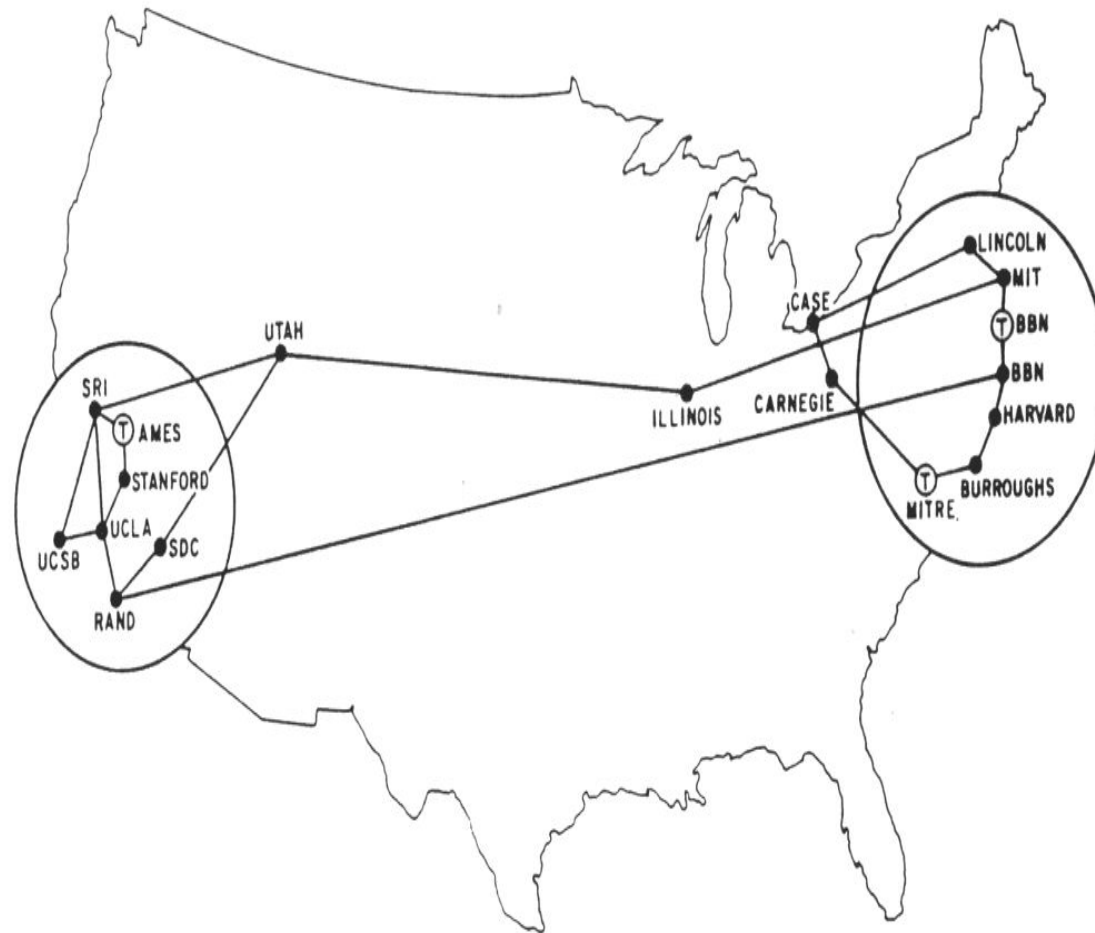
THE ARPA NETWORK

DEC 1969

4 NODES

WHEN

1971

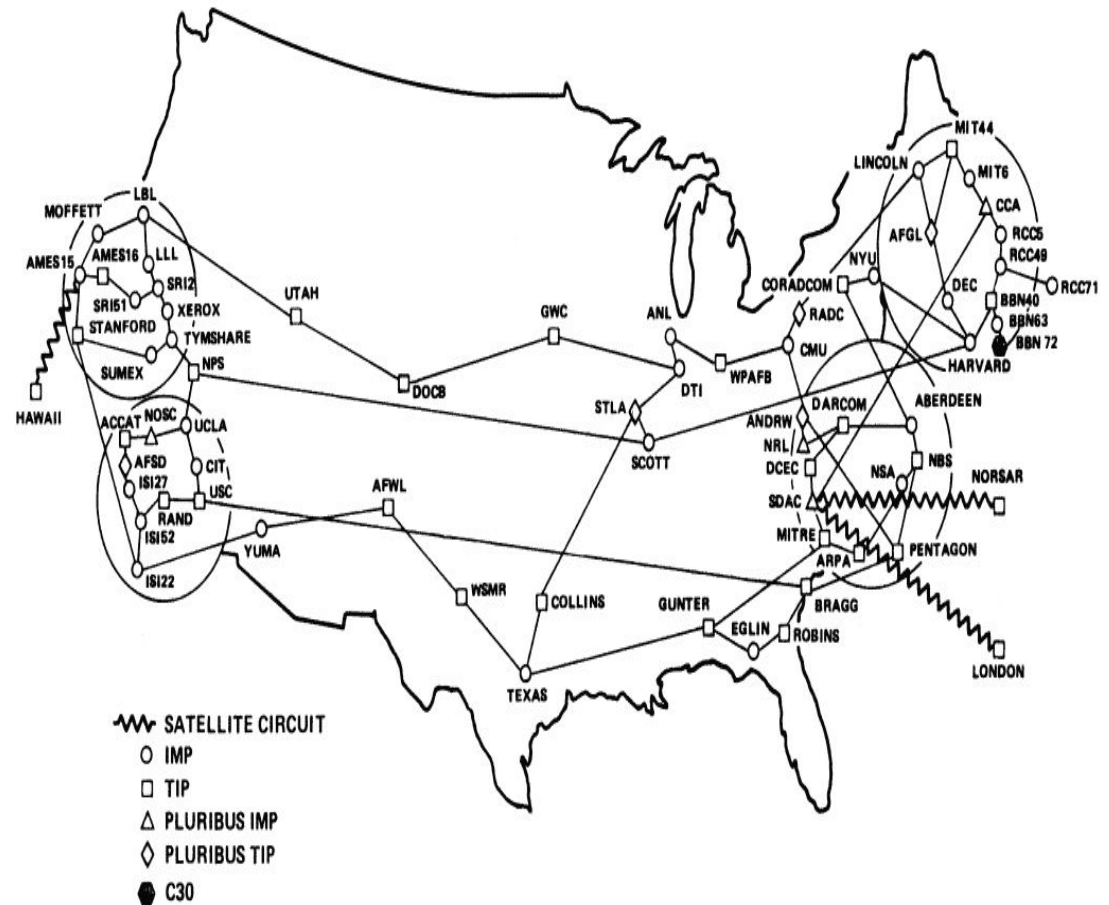


MAP 4 September 1971

WHEN

1980

ARPANET GEOGRAPHIC MAP, OCTOBER 1980

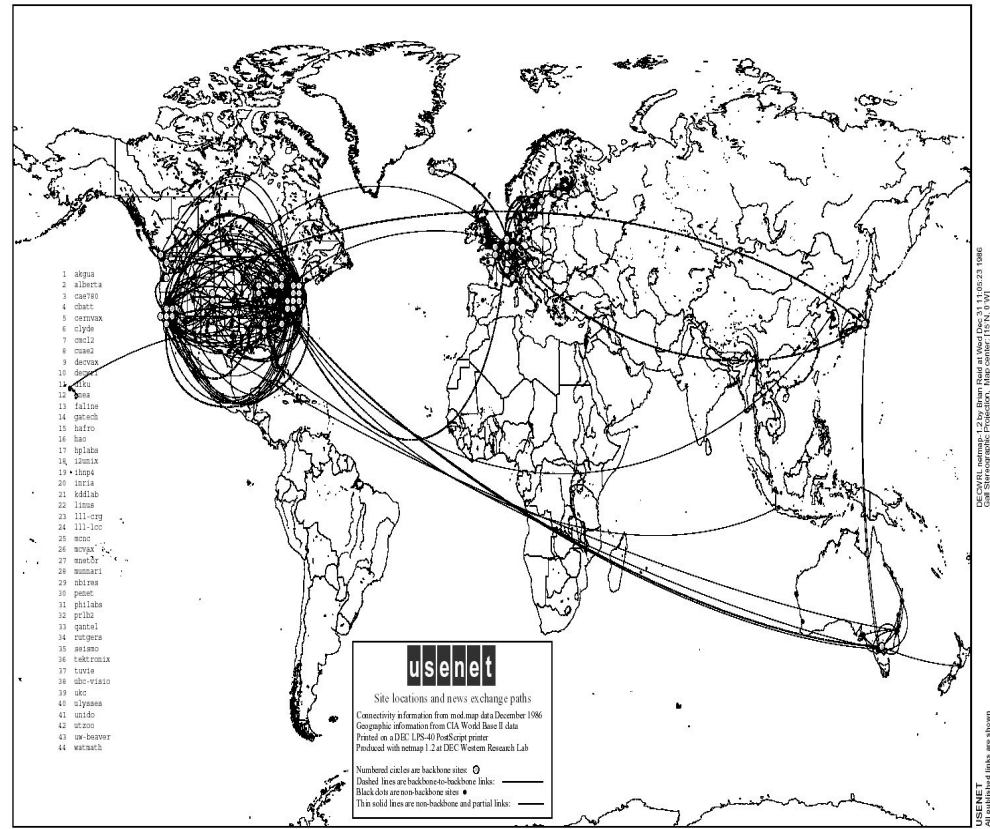


(NOTE: THIS MAP DOES NOT SHOW ARPA'S EXPERIMENTAL SATELLITE CONNECTIONS)

NAMES SHOWN ARE IMP NAMES, NOT (NECESSARILY) HOST NAMES

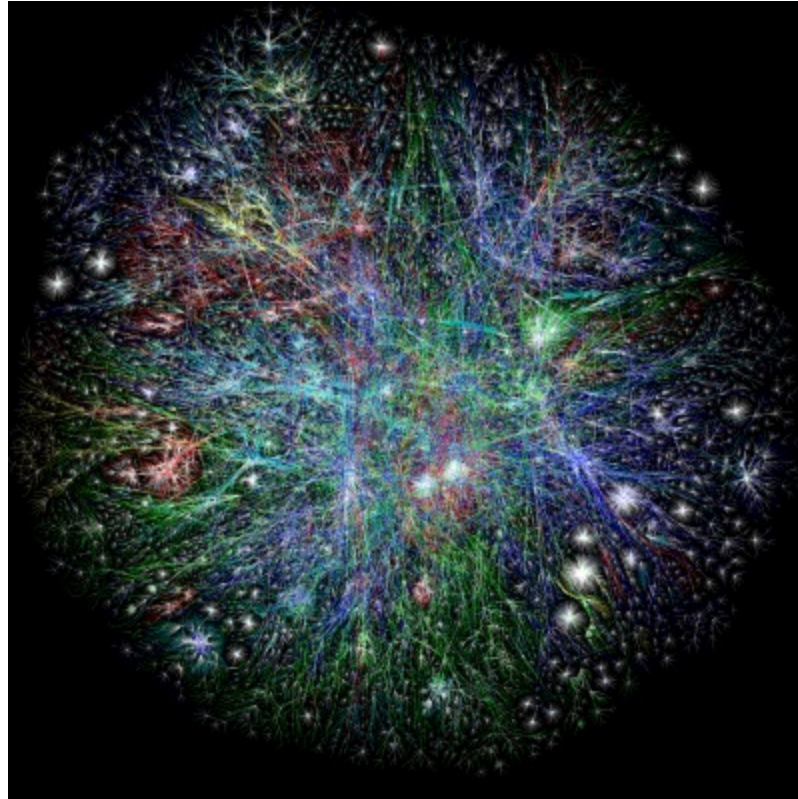
WHEN

1986



WHEN

2006



More Applications

Beyond Internet
Mobile (smart-)phones
Wi-fi
Home Internet
Public Hotspots
Radio / TV Broadcast
Cable TV
Satellite
Personal Networks
Bluetooth
Healthcare / Fitness
Home Networks
Mediacenter

Uses of CN

Computer Networks Vs Distributed Systems

- Uses Of CN:

N/W for companies: RS

High Reliability

Saving Money

Scalability

Powerful Comm. medium

Uses Of CN:

- N/W for People

Access to remote info.

Person to person comm.

Interactive Entertainment

Electronic Commerce

- Social Issues

Social , ethical, political

Home Network Applications

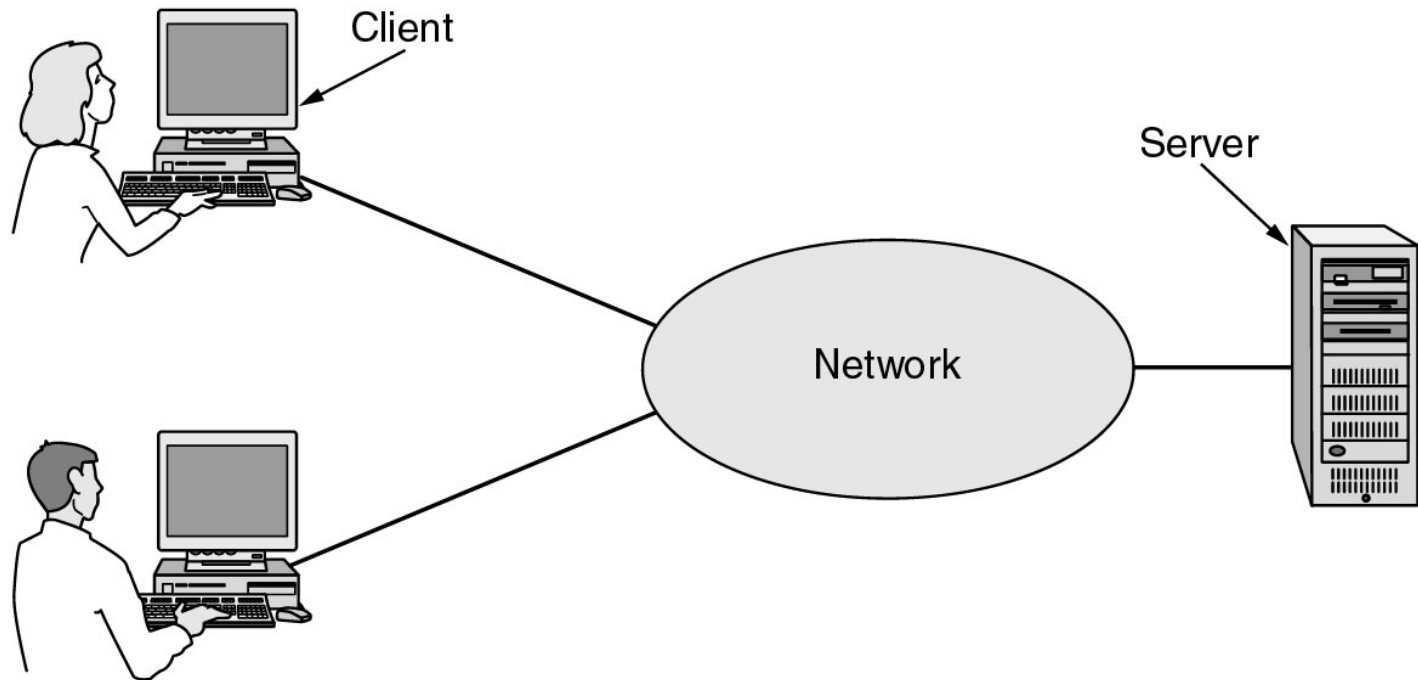
Tag	Full name	Example
B2C	Business-to-consumer	Ordering books on-line
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products on-line
P2P	Peer-to-peer	File sharing

Some forms of e-commerce.

Home Network Applications

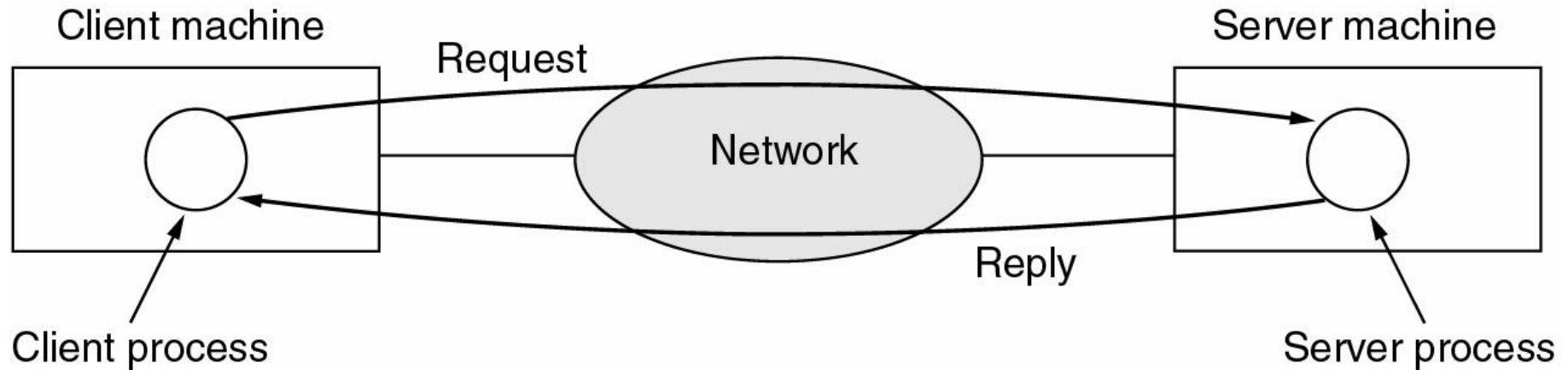
- Access to remote information
- Person-to-person communication
- Interactive entertainment
- Electronic commerce

Business Applications of Networks



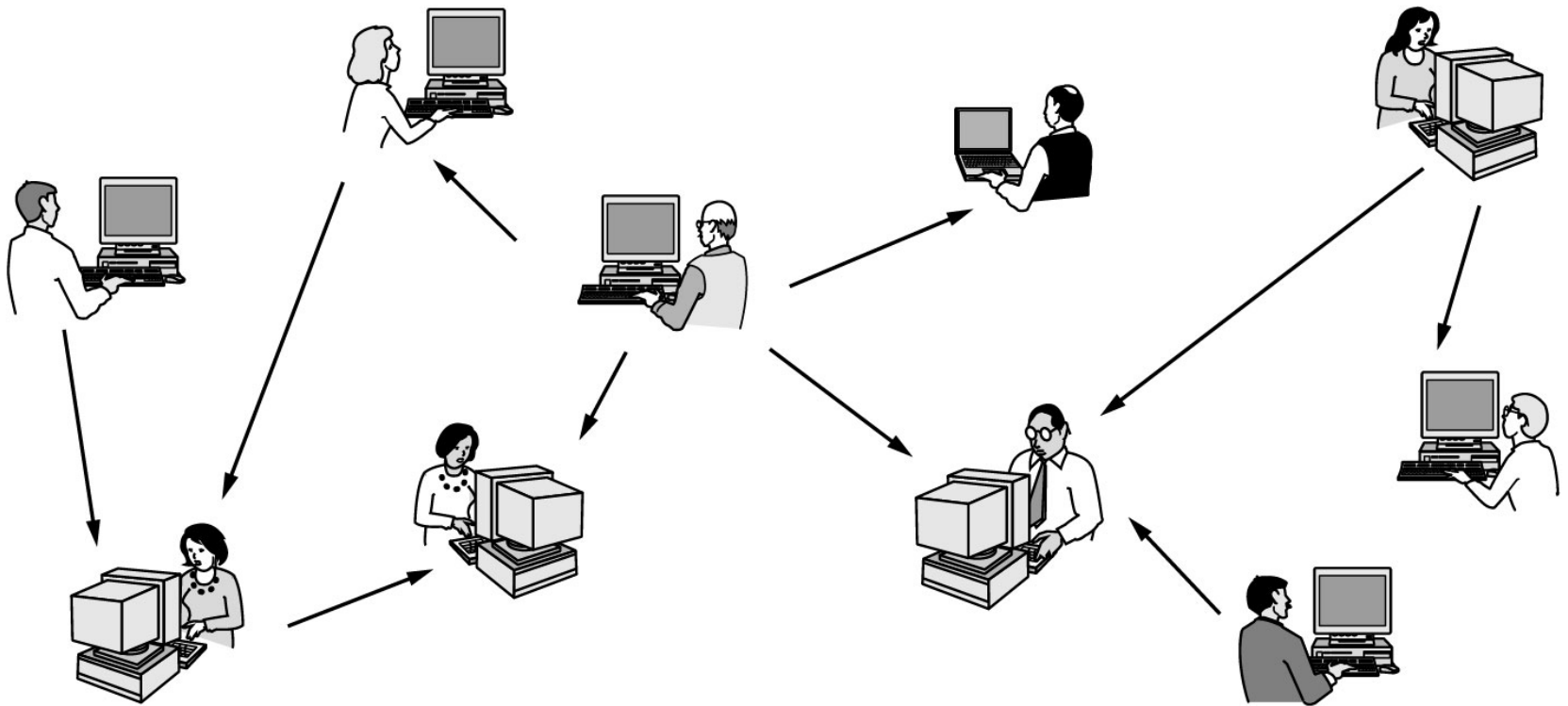
A network with two clients and one server.

Business Applications of Networks (2)



The client-server model involves requests and replies.

Home Network Applications (2)



In peer-to-peer system there are no fixed clients and servers.

Mobile Network Users

Wireless	Mobile	Applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in older, unwired buildings
Yes	Yes	Portable office; PDA for store inventory

Combinations of wireless networks and mobile computing.

Network Hardware

- Local Area Networks
- Metropolitan Area Networks
- Wide Area Networks
- Wireless Networks
- Home Networks
- Internetworks

Broadcast Networks

Types of transmission technology

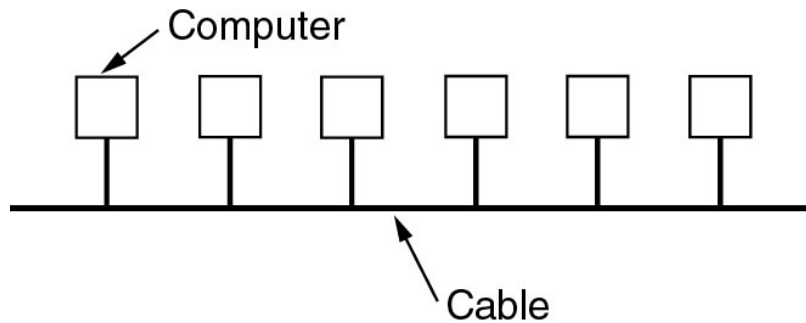
- Broadcast links
- Point-to-point links

Broadcast Networks (2)

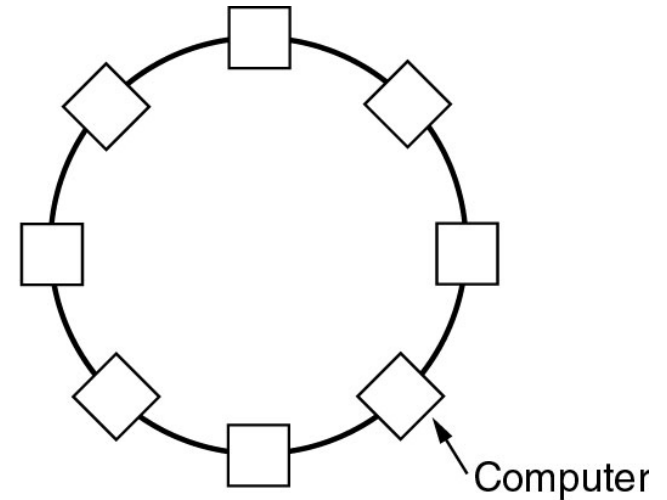
Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	Local area network
100 m	Building	
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Classification of interconnected processors by scale.

Local Area Networks



(a)



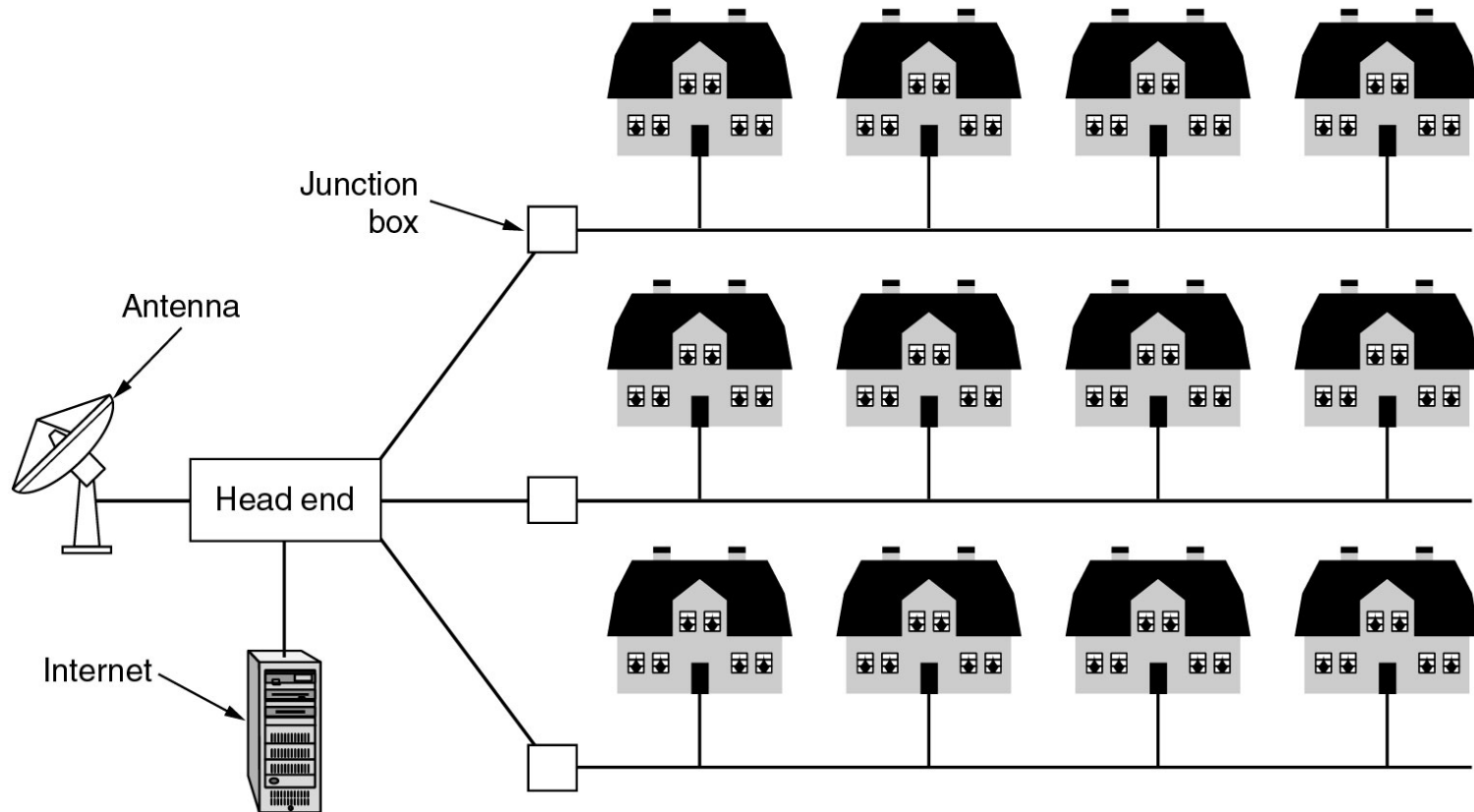
(b)

Two broadcast networks

(a) Bus

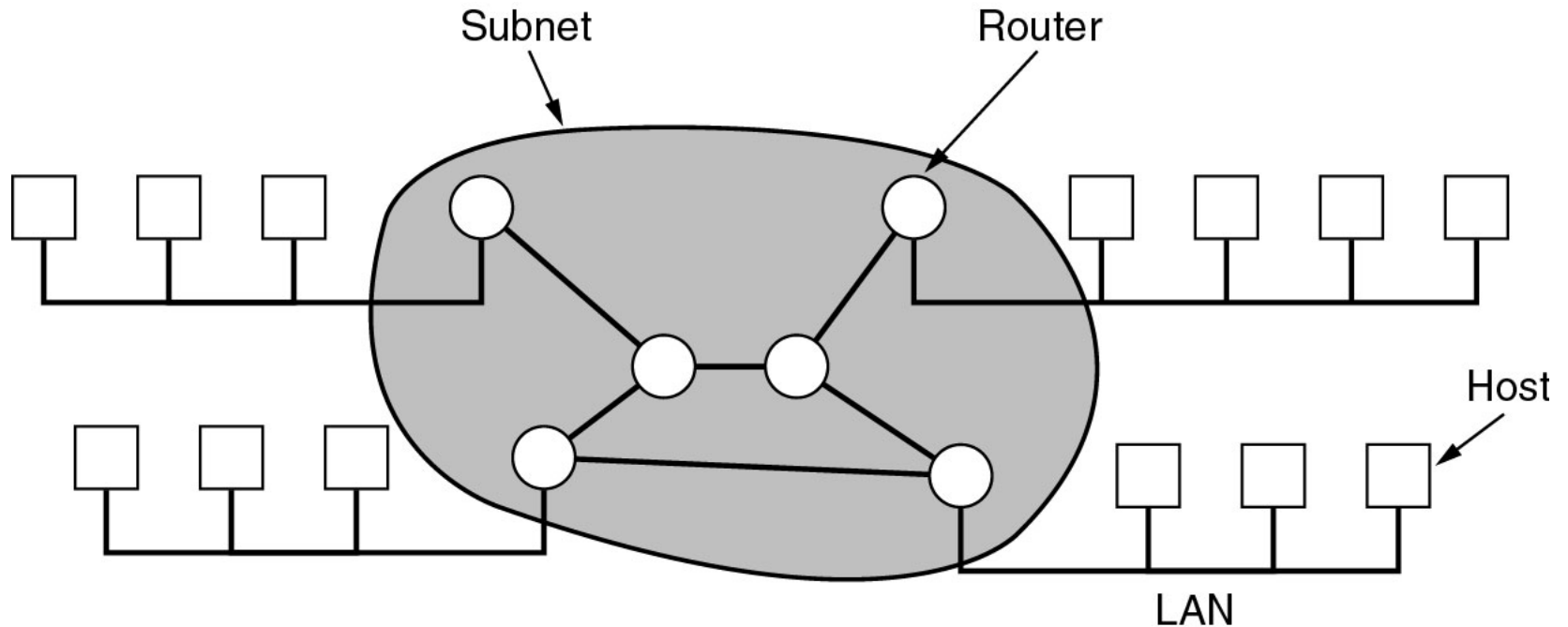
(b) Ring

Metropolitan Area Networks



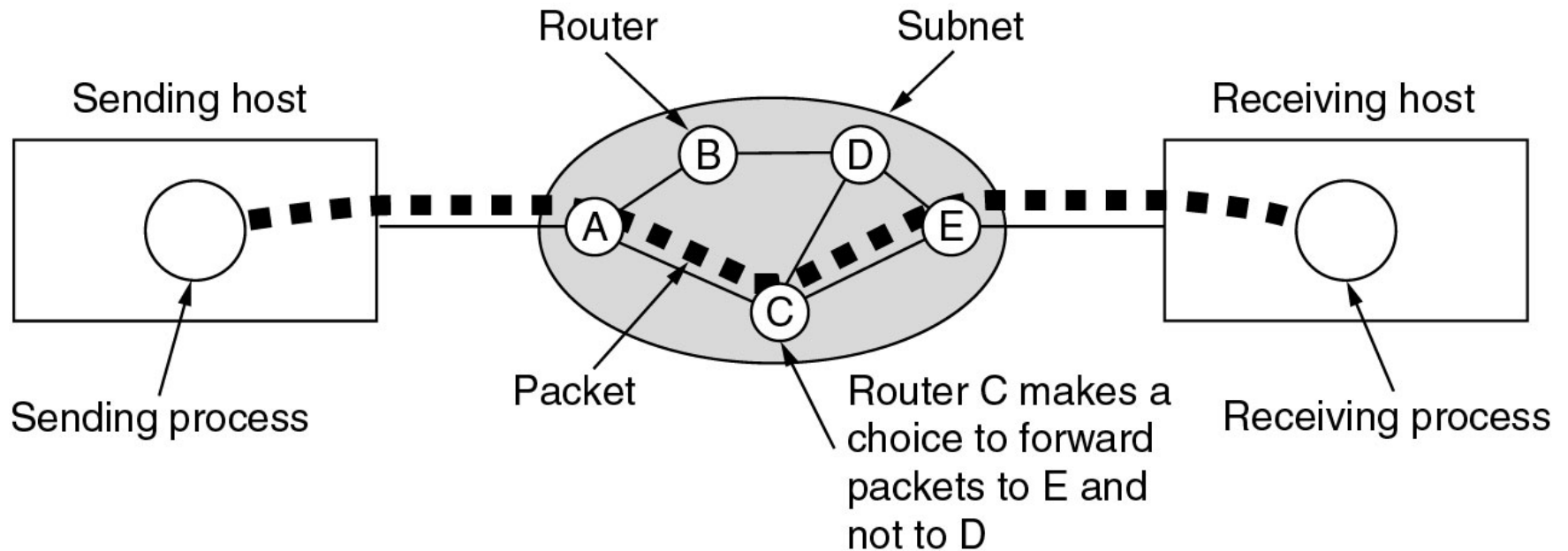
A metropolitan area network based on cable TV.

Wide Area Networks



Relation between hosts on LANs and the subnet.

Wide Area Networks (2)



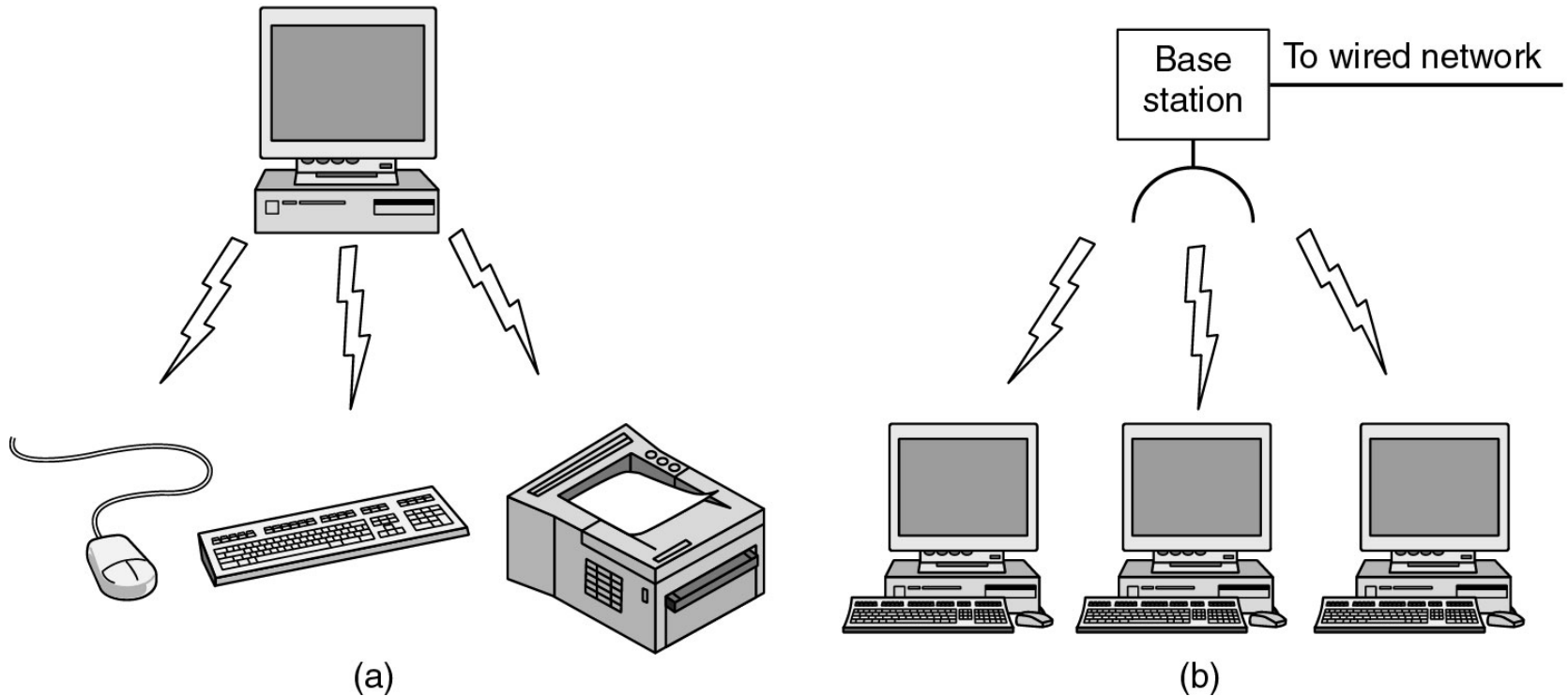
A stream of packets from sender to receiver.

Wireless Networks

Categories of wireless networks:

- System interconnection
- Wireless LANs
- Wireless WANs

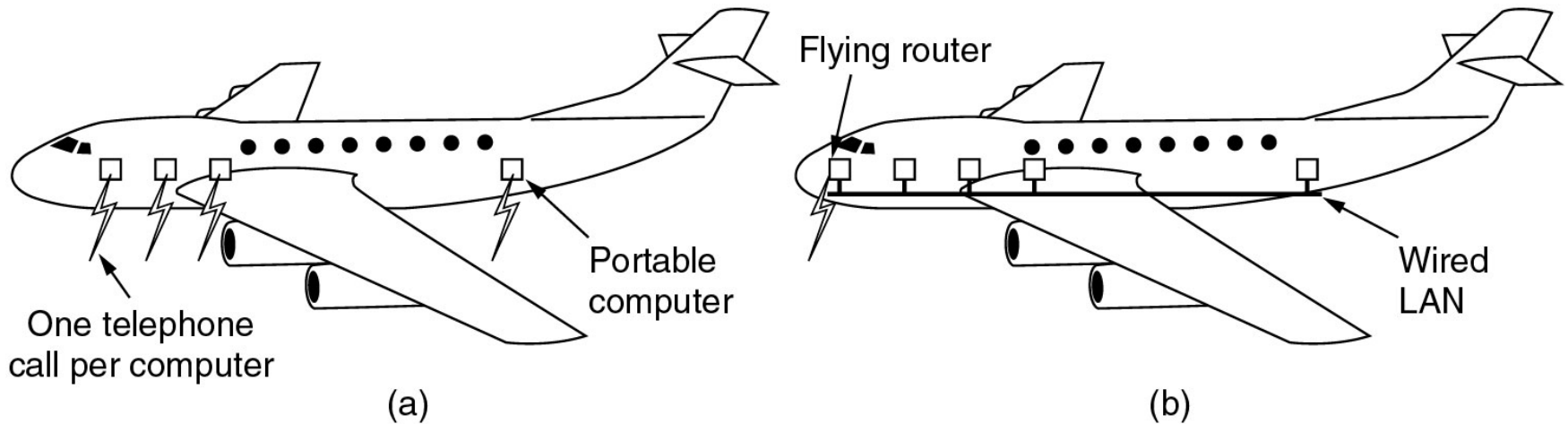
Wireless Networks (2)



(a) Bluetooth configuration

(b) Wireless LAN

Wireless Networks (3)



(a) Individual mobile computers

(b) A flying LAN

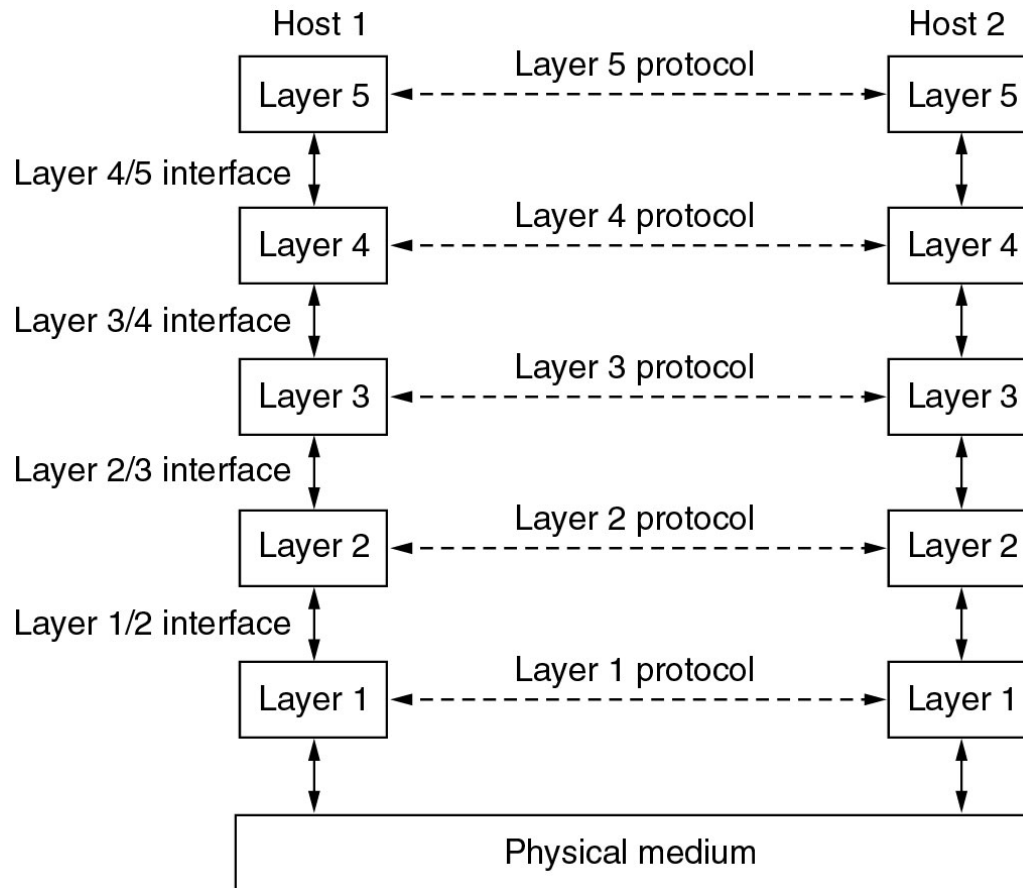
Home Network Categories

- Computers (desktop PC, PDA, shared peripherals)
- Entertainment (TV, DVD, VCR, camera, stereo, MP3)
- Telecomm (telephone, cell phone, intercom, fax)
- Appliances (microwave, fridge, clock, furnace, airco)
- Telemetry (utility meter, burglar alarm, babycam).

Network Software

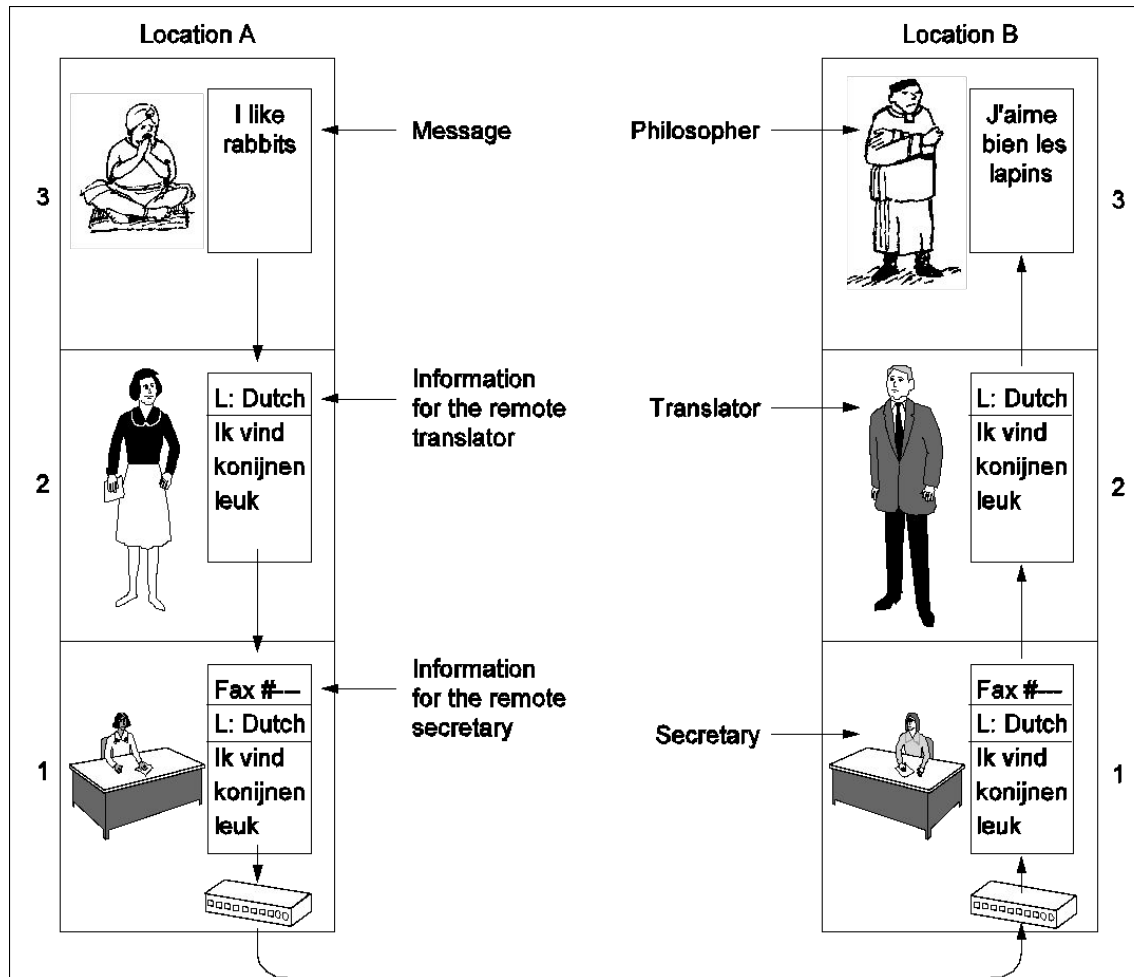
- Protocol Hierarchies
- Design Issues for the Layers
- Connection-Oriented and Connectionless Services
- Service Primitives
- The Relationship of Services to Protocols

Network Software Protocol Hierarchies



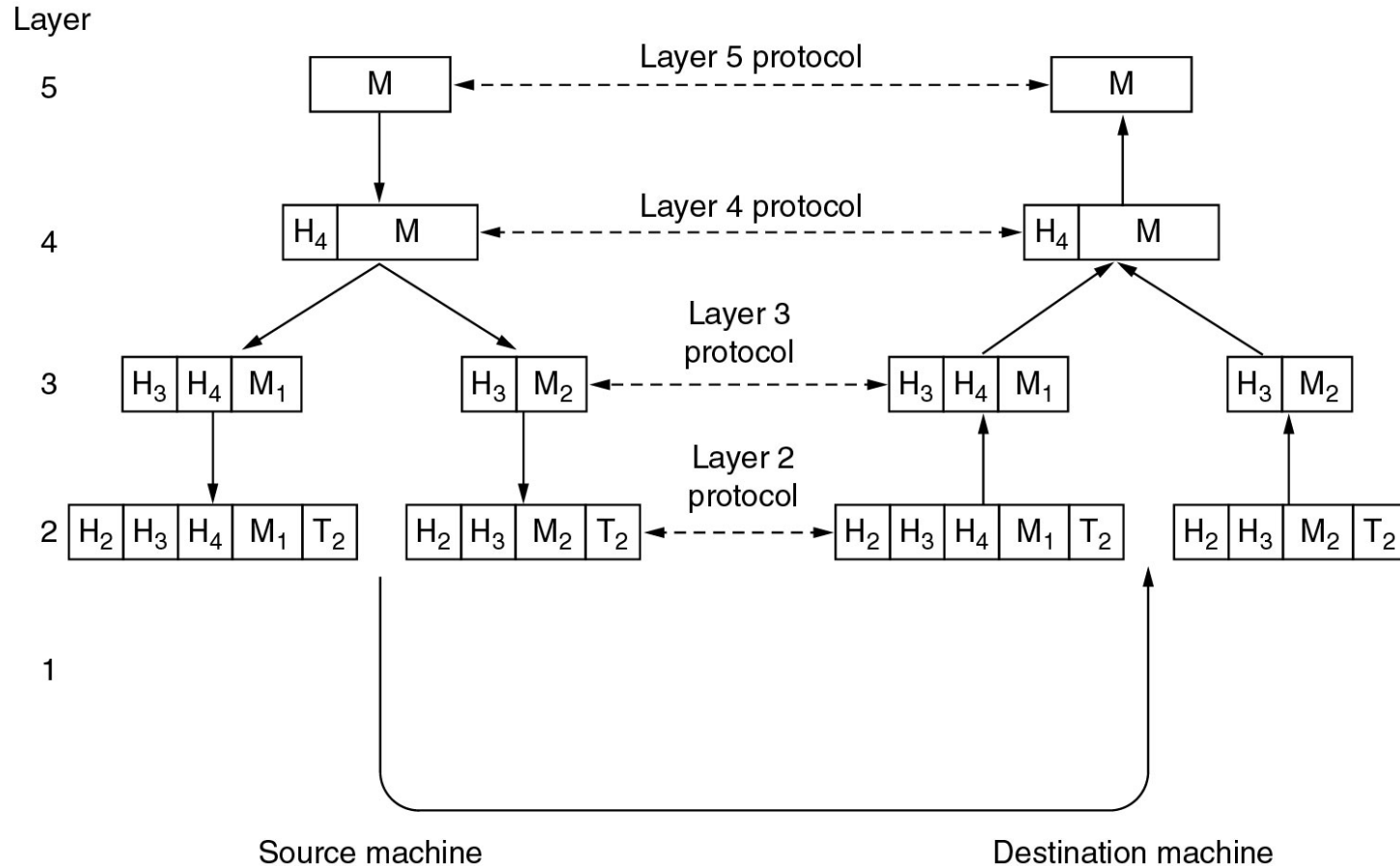
Layers, protocols, and interfaces.

Protocol Hierarchies (2)



The philosopher-translator-secretary architecture.

Protocol Hierarchies (3)

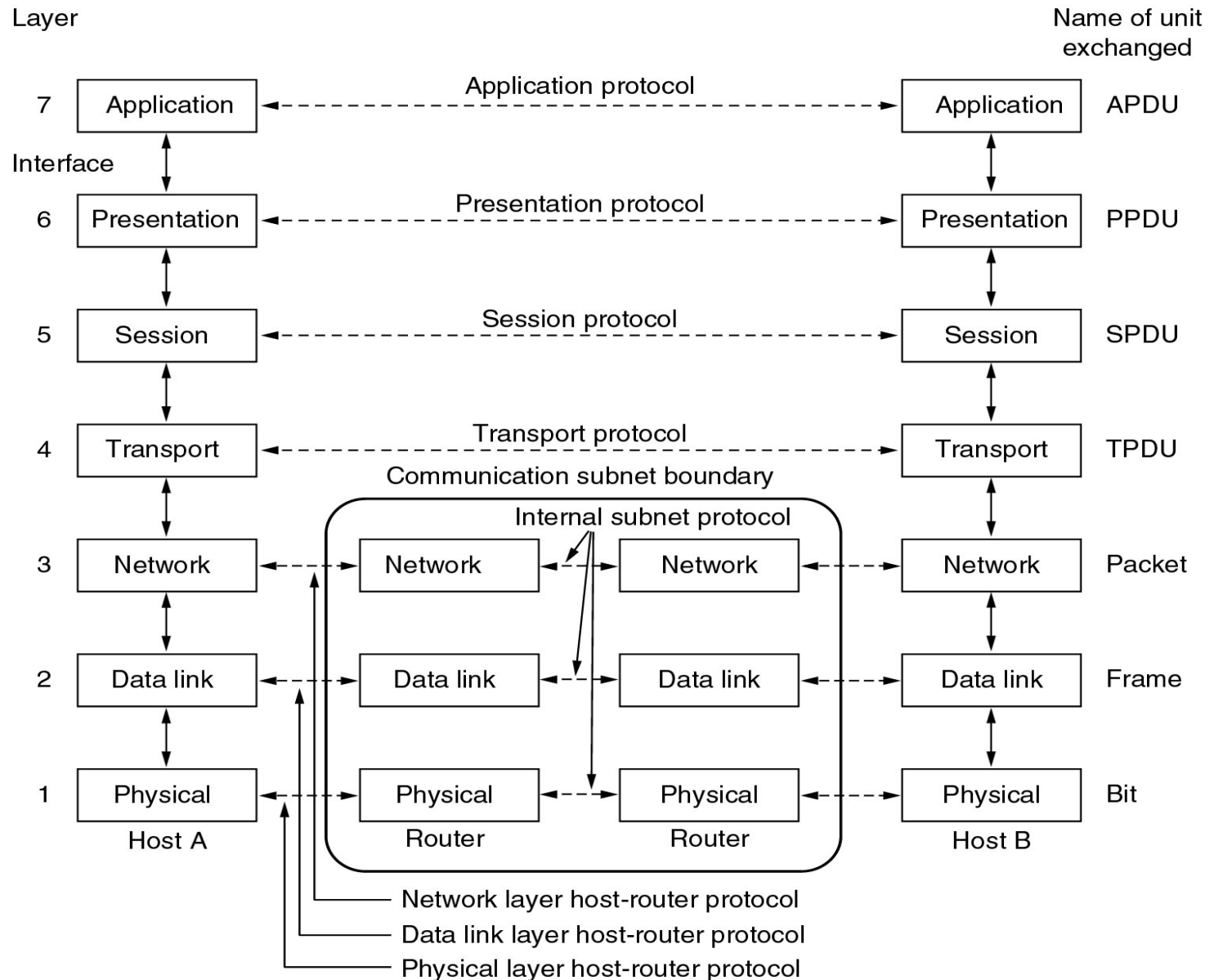


Example information flow supporting virtual communication in layer 5.

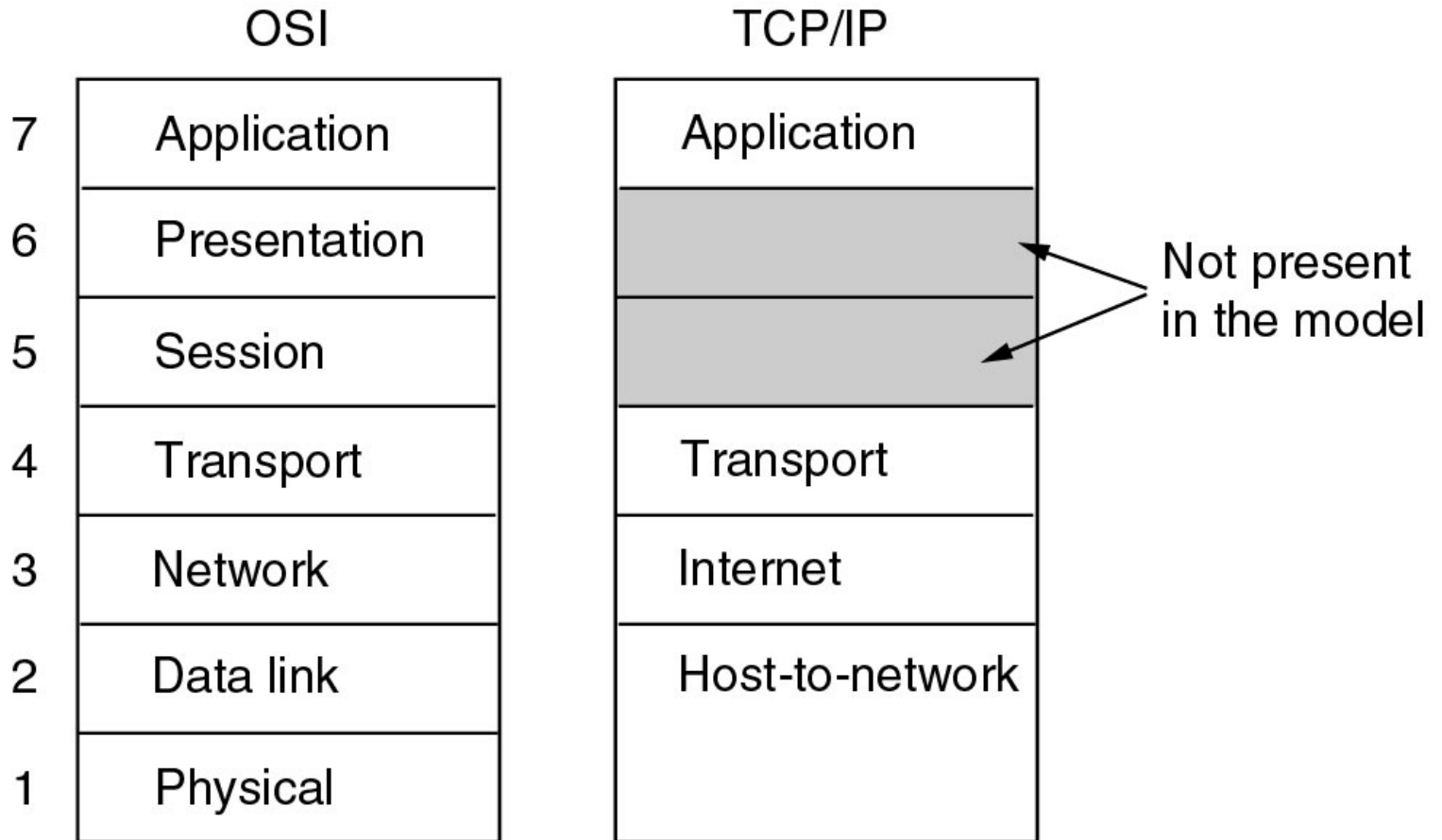
Design Issues for the Layers

- Addressing
- Error Control
- Flow Control
- Multiplexing
- Routing

The OSI reference model.

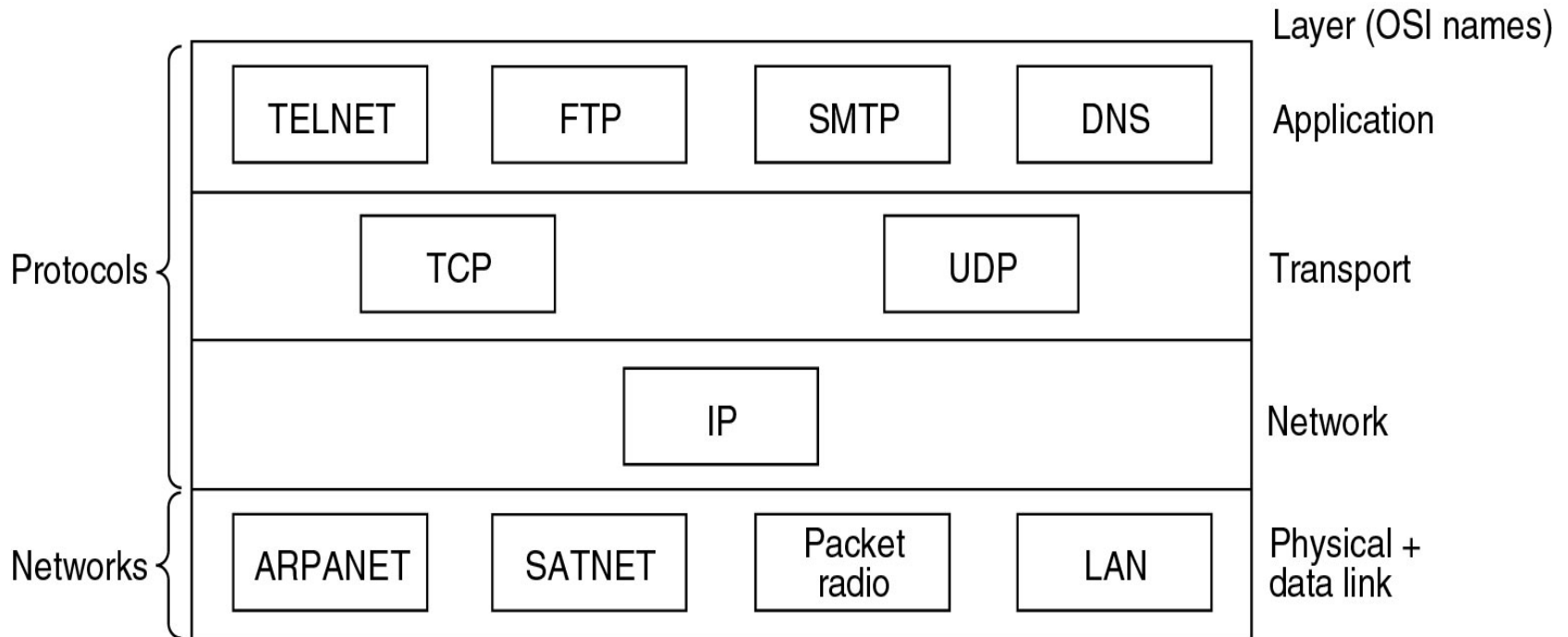


Reference Models (2)



The TCP/IP reference model.

Reference Models (3)



Protocols and networks in the TCP/IP model initially.

Comparing OSI and TCP/IP Models

Concepts central to the OSI model

- Services
- Interfaces
- Protocols